

Motor Predictive Maintenance

Developed a Machine Learning application to predict motor failures using sensor data.

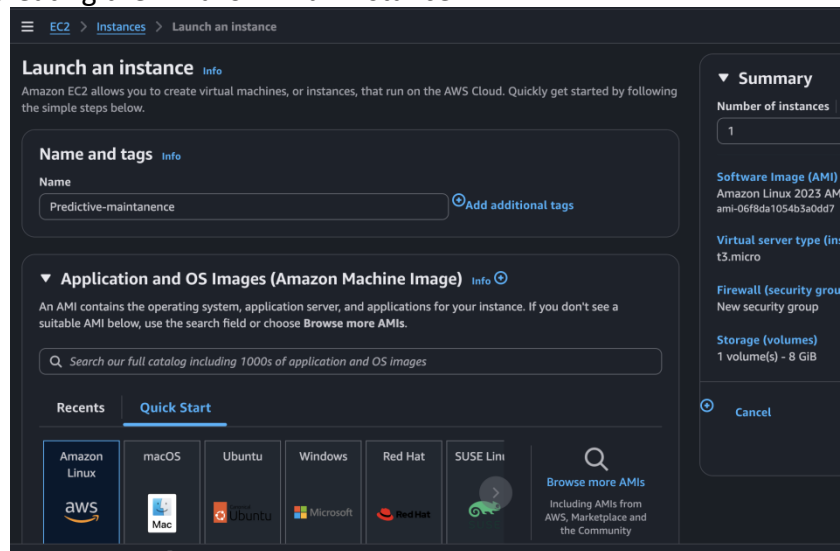
Deployed and hosted the Streamlit application on an AWS EC2 instance running Amazon Linux.

Configured the server and accessed the application through the EC2 public IP.

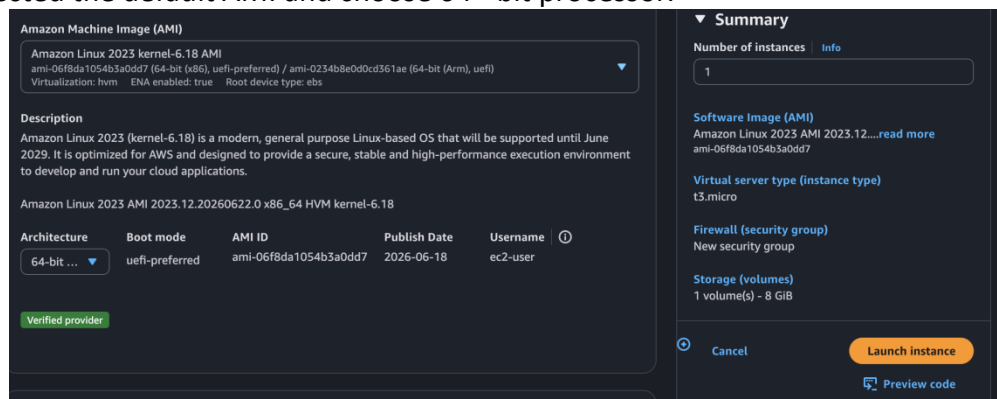
Technologies: Python, Streamlit, Scikit-Learn, AWS EC2, Amazon Linux, Git

Repository: https://github.com/Derick-Roshan/Predictive_maintanance

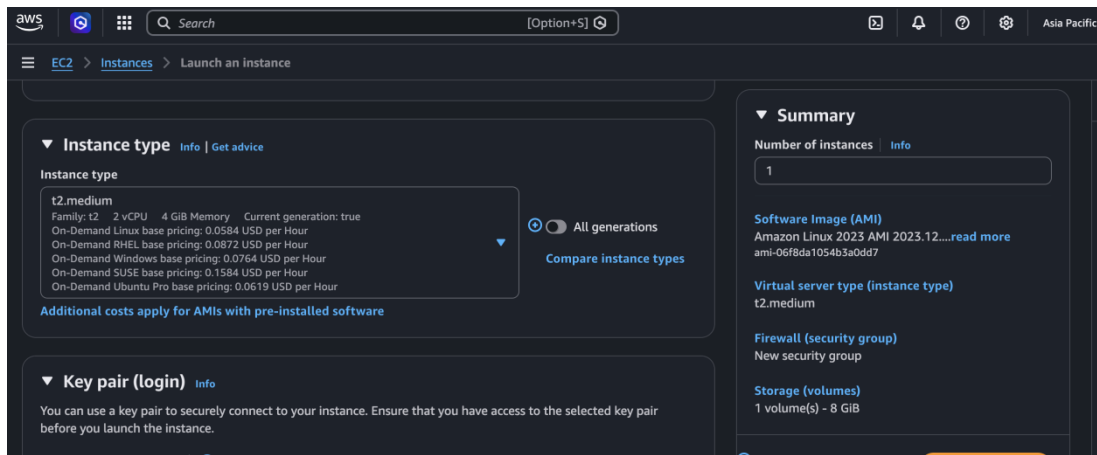
Step 1 - Creating the Amazon Linux instance.



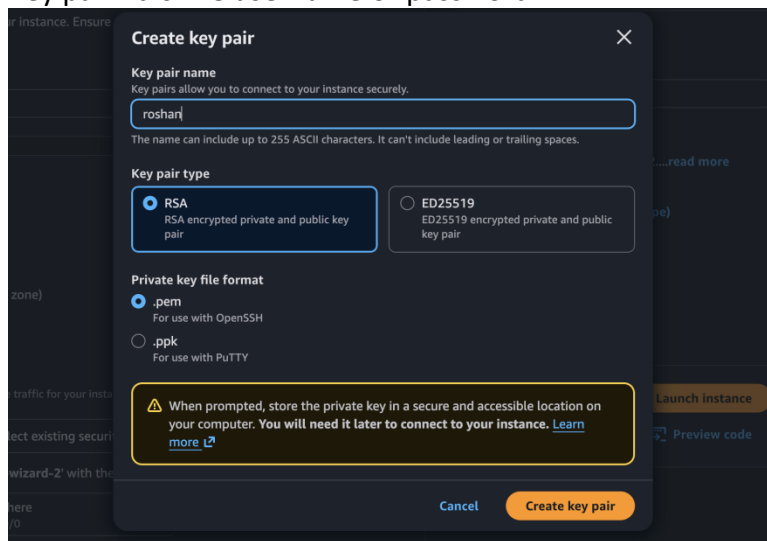
Selected the default AMI and choose 64 - bit processor.



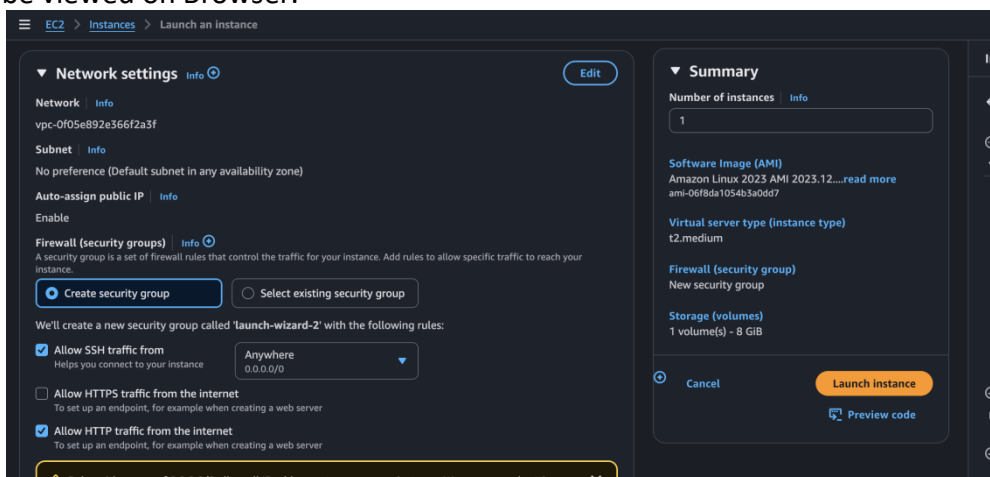
Selecting the t2.medium, while running the Application instance type must be huge.



Creating the Key pair. It is like username or password.



Choose the Default VPC and Security Group & Turned on the HTTP traffic then only it can be viewed on Browser.



Configured the Storage as 30GB.

Configure storage info Advanced

1x GiB Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

☐ S3 Files - new ☐ EFS

☐ FSx ☒ None

Number of instances info

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-06f8da1054b3a0dd7

Virtual server type (instance type)
t2.medium

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Number of instance is 1. and click on Launch instance.

Summary

Number of instances Info

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....[read more](#)
ami-06f8da1054b3a0dd7

Virtual server type (instance type)
t2.medium

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

You can see that Instance is running.

EC2 > Instances

Instances (2) Info [Connect](#) [Instance state](#) [Actions](#)

Saved filter sets
Choose filter set

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public
☐	linux	i-0aa9299d9ddce00a	Stopped	t2.medium	-	ap-southeast-2a	-
☐	Predictive-mainta...	i-09f16ab363f3e1531	Running	t2.medium	Initializing	ap-southeast-2a	ec2-15

I copied a IP address for connecting the Linux server.

Instance summary for i-09f16ab36

Updated less than a minute ago

Instance ID
 i-09f16ab36f3e1531

IPv6 address
 -

Hostname type
 IP name: ip-172-31-1-91.ap-southeast-2.compute.internal

 **Public IPv4 address copied**

 15.134.129.188 | [open address ↗](#)

Instance state
 **Running**

Private IP DNS name (IPv4 only)
 ip-172-31-1-91.ap-southeast-2.compute.internal

Maintenance [Info](#)

 [Connect](#) [Instance state ▼](#) [Actions ▼](#)

Private IPv4 addresses
 172.31.1.91

Public DNS
 ec2-15-134-129-188.ap-southeast-2.compute.amazonaws.com | [open address ↗](#)

```
r@Roshans-MacBook-Pro ~ % cd Downloads
r@Roshans-MacBook-Pro Downloads % chmod 400 roshan.pem
r@Roshans-MacBook-Pro Downloads % ssh -i roshan.pem ec2-user@15.134.129.188
The authenticity of host '15.134.129.188 (15.134.129.188)' can't be established.
ED25519 key fingerprint is: SHA256:Geudas92mAnf/RxFyJDEt9Xv8SrdxPzfS9qcGMswBAQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '15.134.129.188' (ED25519) to the list of known hosts.
```



```
#_
~\#### Amazon Linux 2023
~~\#####
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
    V~! -->
      / \
     /   \
    /m/'   \
```

```
[ec2-user@ip-172-31-1-91 ~]$ █
```

We need to clone the project from the github repo.

Then we need to download the packages and libraries which we used in the python.

```
[root@ip-172-31-1-91 Predictive_maintenance]# python3 predictive_maintenance_app.py
Traceback (most recent call last):
  File "/home/ec2-user/Predictive_maintenance/predictive_maintenance_app.py", line 1, in <module>
    import streamlit as st
ModuleNotFoundError: No module named 'streamlit'
[root@ip-172-31-1-91 Predictive_maintenance]#
```

```
Found existing installation: python-dateutil 2.8.1
Uninstalling python-dateutil-2.8.1:
  Successfully uninstalled python-dateutil-2.8.1
Attempting uninstall: requests
Found existing installation: requests 2.25.1
ERROR: Cannot uninstall requests 2.25.1, RECORD file not found. Hint: The package was installed by rpm.
[root@ip-172-31-1-91 Predictive_maintenance]# python3 -m venv venv
[root@ip-172-31-1-91 Predictive_maintenance]# source venv/bin/activate
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# pip install --upgrade pip
Requirement already satisfied: pip in ./venv/lib/python3.9/site-packages (21.3.1)
Collecting pip
  Downloading pip-26.0.1-py3-none-any.whl (1.8 MB)
    |#####| 1.8 MB 14.3 MB/s
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 21.3.1
    Uninstalling pip-21.3.1:
      Successfully uninstalled pip-21.3.1
Successfully installed pip-26.0.1
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# pip install streamlit
```

```
Downloading pyarrow-21.0.0-cp39-cp39-manylinux_2_28_x86_64.whl (42.7 MB)
  42.7/42.7 MB 119.4 MB/s 0:00:00
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Downloading pytz-2026.2-py2.py3-none-any.whl (510 kB)
Downloading referencing-0.36.2-py3-none-any.whl (26 kB)
Downloading rpdspy-0.27.1-cp39-cp39-manylinux_2_17_x86_64_manylinux2014_x86_64.whl (384 kB)
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: pytz, watchdog, urllib3, tzdata, typing-extensions, tornado, toml, tenacity, smmap, six, r
pds-py, pyarrow, protobuf, pillow, packaging, numpy, narwhals, MarkupSafe, idna, click, charset-normalizer, certifi, cach
etools, blinker, attrs, requests, referencing, python-dateutil, jinja2, gitdb, pydeck, pandas, jsonschema-specifications,
gitpython, jsonschema, altair, streamlit
Successfully installed MarkupSafe-3.0.3 altair-5.5.0 attrs-26.1.0 blinker-1.9.0 cachetools-6.2.6 certifi-2026.6.17 charse
t-normalizer-3.4.7 click-8.1.8 gitdb-4.0.12 gitpython-3.1.50 idna-3.18 jinja2-3.1.6 jsonschema-4.25.1 jsonschema-specific
ations-2025.9.1 narwhals-2.21.0 numpy-2.0.2 packaging-25.0 pandas-2.3.3 pillow-11.3.0 protobuf-6.33.6 pyarrow-21.0.0 pyde
ck-0.9.2 python-dateutil-2.9.0.post0 pytz-2026.2 referencing-0.36.2 requests-2.32.5 rpdspy-0.27.1 six-1.17.0 smmap-5.0.3
streamlit-1.50.0 tenacity-9.1.2 toml-0.10.2 tornado-6.5.7 typing-extensions-4.15.0 tzdata-2026.2 urllib3-2.6.3 watchdog-
6.0.0
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# python3 predictive_maintenance_app.py
Traceback (most recent call last):
  File "/home/ec2-user/Predictive_maintenance/predictive_maintenance_app.py", line 4, in <module>
    from sklearn.preprocessing import StandardScaler
ModuleNotFoundError: No module named 'sklearn'
(venv) [root@ip-172-31-1-91 Predictive_maintenance]#
```

```
Downloading toolz-1.1.0-py3-none-any.whl (58 kB)
Downloading zipp-3.23.1-py3-none-any.whl (10 kB)
Downloading lock-1.0.0-py2.py3-none-any.whl (4.4 kB)
Downloading scikit-learn-1.6.1-cp39-cp39-manylinux_2_17_x86_64_manylinux2014_x86_64.whl (13.5 MB)
  13.5/13.5 MB 123.8 MB/s 0:00:00
Downloading joblib-1.5.3-py3-none-any.whl (309 kB)
Downloading scipy-1.13.1-cp39-cp39-manylinux_2_17_x86_64_manylinux2014_x86_64.whl (38.6 MB)
  38.6/38.6 MB 107.5 MB/s 0:00:00
Downloading threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Building wheels for collected packages: StandardScaler
  Building wheel for StandardScaler (pyproject.toml) ... done
  Created wheel for StandardScaler: filename=standardscaler-0.5-py3-none-any.whl size=2653 sha256=94a4c666c2e7823506e2e93
584fe6279d88ec74abc1a91f9f2bfb0444f39890d
  Stored in directory: /root/.cache/pip/wheels/98/7d/ec/7ea52e96c7cf25f808c75a53cb1db1ef8e7c071debb983276
Successfully built StandardScaler
Installing collected packages: zipp, toolz, threadpoolctl, scipy, pyyaml, lock, joblib, fsspec, cloudpickle, scikit-learn,
partd, importlib-metadata, scikit-elm, dask, StandardScaler
Successfully installed StandardScaler-0.5 cloudpickle-3.1.2 dask-2024.8.0 fsspec-2025.10.0 importlib-metadata-8.7.1 jobli
b-1.5.3 lock-1.0.0 partd-1.4.2 pyyaml-6.0.3 scikit-learn-1.6.1 scikit-learn-1.6.1 scipy-1.13.1 threadpoolctl-3.6.0 toolz
-1.1.0 zipp-3.23.1
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# python3 predictive_maintenance_app.py
Traceback (most recent call last):
  File "/home/ec2-user/Predictive_maintenance/predictive_maintenance_app.py", line 5, in <module>
    from xgboost import XGBClassifier
ModuleNotFoundError: No module named 'xgboost'
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# pip install xgboost
```



```

Traceback (most recent call last):
  File "/home/ec2-user/Predictive_maintenance/predictive_maintenance_app.py", line 5, in <module>
    from xgboost import XGBClassifier
ModuleNotFoundError: No module named 'xgboost'
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# pip install xgboost
Collecting xgboost
  Downloading xgboost-2.1.4-py3-none-manylinux_2_28_x86_64.whl.metadata (2.1 kB)
Requirement already satisfied: numpy in ./env/lib/python3.9/site-packages (from xgboost) (2.0.2)
Collecting nvidia-nccl-cu12 (from xgboost)
  Downloading nvidia_nccl_cu12-2.30.7-py3-none-manylinux_2_18_x86_64.whl.metadata (2.1 kB)
Requirement already satisfied: scipy in ./env/lib/python3.9/site-packages (from xgboost) (1.13.1)
Downloading xgboost-2.1.4-py3-none-manylinux_2_28_x86_64.whl (223.6 MB)
 223.6/223.6 MB 50.7 MB/s 0:00:04
Downloading nvidia_nccl_cu12-2.30.7-py3-none-manylinux_2_18_x86_64.whl (303.4 MB)
 303.4/303.4 MB 41.7 MB/s 0:00:06
Installing collected packages: nvidia-nccl-cu12, xgboost
Successfully installed nvidia-nccl-cu12-2.30.7 xgboost-2.1.4
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# python3 predictive_maintenance_app.py
Traceback (most recent call last):
  File "/home/ec2-user/Predictive_maintenance/predictive_maintenance_app.py", line 8, in <module>
    import matplotlib.pyplot as plt
ModuleNotFoundError: No module named 'matplotlib'
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# pip install matplotlib

```

```

(venv) [root@ip-172-31-1-91 Predictive_maintenance]# python3 predictive_maintenance_app.py
Traceback (most recent call last):
  File "/home/ec2-user/Predictive_maintenance/predictive_maintenance_app.py", line 9, in <module>
    import seaborn as sns
ModuleNotFoundError: No module named 'seaborn'
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# pip install seaborn

```

After installing the python packages and if we need to run the project you can see that is running in the bare mode.

```

(venv) [root@ip-172-31-1-91 Predictive_maintenance]# python3 predictive_maintenance_app.py
2026-06-23 17:14:41.875 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:41.875 WARNING streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.018
Warning: to view this Streamlit app on a browser, run it with the following
command:
    streamlit run predictive_maintenance_app.py [ARGUMENTS]
2026-06-23 17:14:42.018 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.018 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.018 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.018 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.018 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.025 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.025 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-06-23 17:14:42.025 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

```

To sort it out:

Now you see that the after using nohup command the project is running in the background.

```

(venv) [root@ip-172-31-1-91 Predictive_maintenance]# nohup streamlit run predictive_maintenance_app.py --server.port 8502 --server.address 0.0.0.0 > streamlit.log 2>&1 &
[1] 29338
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# ps-aux | grep streamlit
bash: ps-aux: command not found
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# nohup streamlit run predictive_maintenance_app.py --server.port 8502 --server.address 0.0.0.0 > streamlit.log 2>&1 &
[2] 29341
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# ps aux | grep streamlit
root      29338  0.8  1.0 268288 43532 pts/1    S   17:28   0:00 /home/ec2-user/Predictive_maintenance/venv/bin/python3 /home/ec2-user/Predictive_maintenance/venv/bin/streamlit run predictive_maintenance_app.py --server.port 8502 --server.address 0.0.0.0
root      29343  0.0  0.0 222344 2176 pts/1    S+  17:28   0:00 grep --color=auto streamlit
[2]+  Exit 1                  nohup streamlit run predictive_maintenance_app.py --server.port 8502 --server.address 0.0.0.0 > streamlit.log 2>&1
(venv) [root@ip-172-31-1-91 Predictive_maintenance]# tail -f streamlit.log
nohup: ignoring input

Collecting usage statistics. To deactivate, set browser.gatherUsageStats to false.

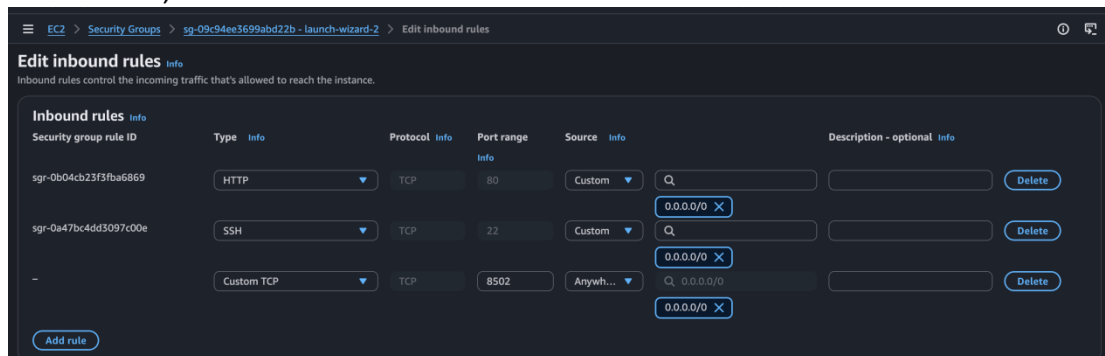
2026-06-23 17:28:35.004 Port 8502 is already in use

```

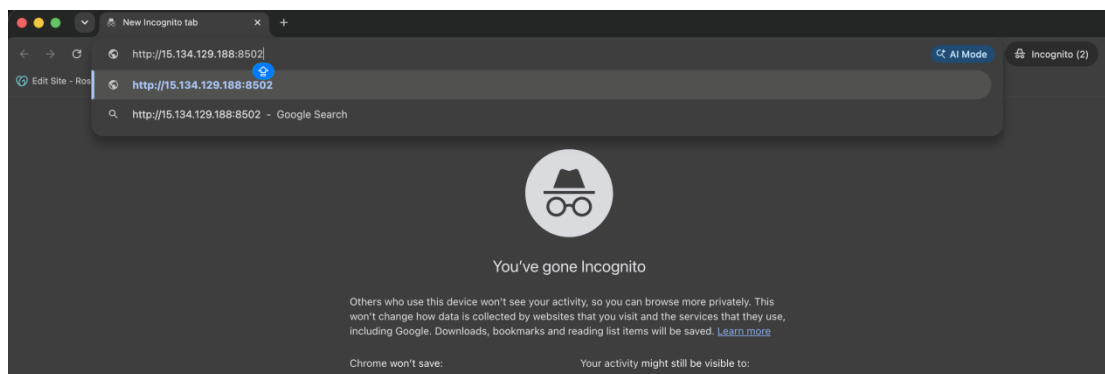
To view the project im using the chrom browser.

[http:// instance](http://instance) ip address: 8502

Before that we need to check wheather inbound rules on Security groups has been added - TCP, Port & Source.



After saving - I entered a IP address and Port number to view the project.



You can see that the project has been deployed Successfully.

